

Spine Cooling Runtime

Brief operator guide · Raspberry Pi medical-device prototype

How to use it

- 1 The device enters **Cooling** by itself. The compressor cools the plate to -1°C .
- 2 Seat the cartridge until **Cartridge In Place** is ON. Fill until **Level Low** and **Level Critical** are OK.
- 3 Place the catheter. Set the CSF target with $+ / -$ ($30\text{--}35^{\circ}\text{C}$).
- 4 Press **START COOLING**. The pump runs closed-loop toward the setpoint.

Screens

Main is the treatment page. Expert is monitoring. Service can move hardware and is one extra tap away on purpose.

Main screen — Ready



START COOLING stays disabled until the cartridge is seated and both level switches are OK.

Main screen — Cooling



The compressor is already cooling the plate. Place the catheter, set the CSF target with + / −, then press START COOLING.

Main screen — Error



A stop fault latches the Error state, stops pumping, and shows the message. Tap the message for help. ACKNOWLEDGE ERROR is enabled only after the fault condition has cleared.

Error help sheet

Level sensor failure detected

Probable causes

- Coolant level has dropped below a level switch.
- The cartridge is not filled enough.
- A level switch is stuck, unplugged, or wired incorrectly.

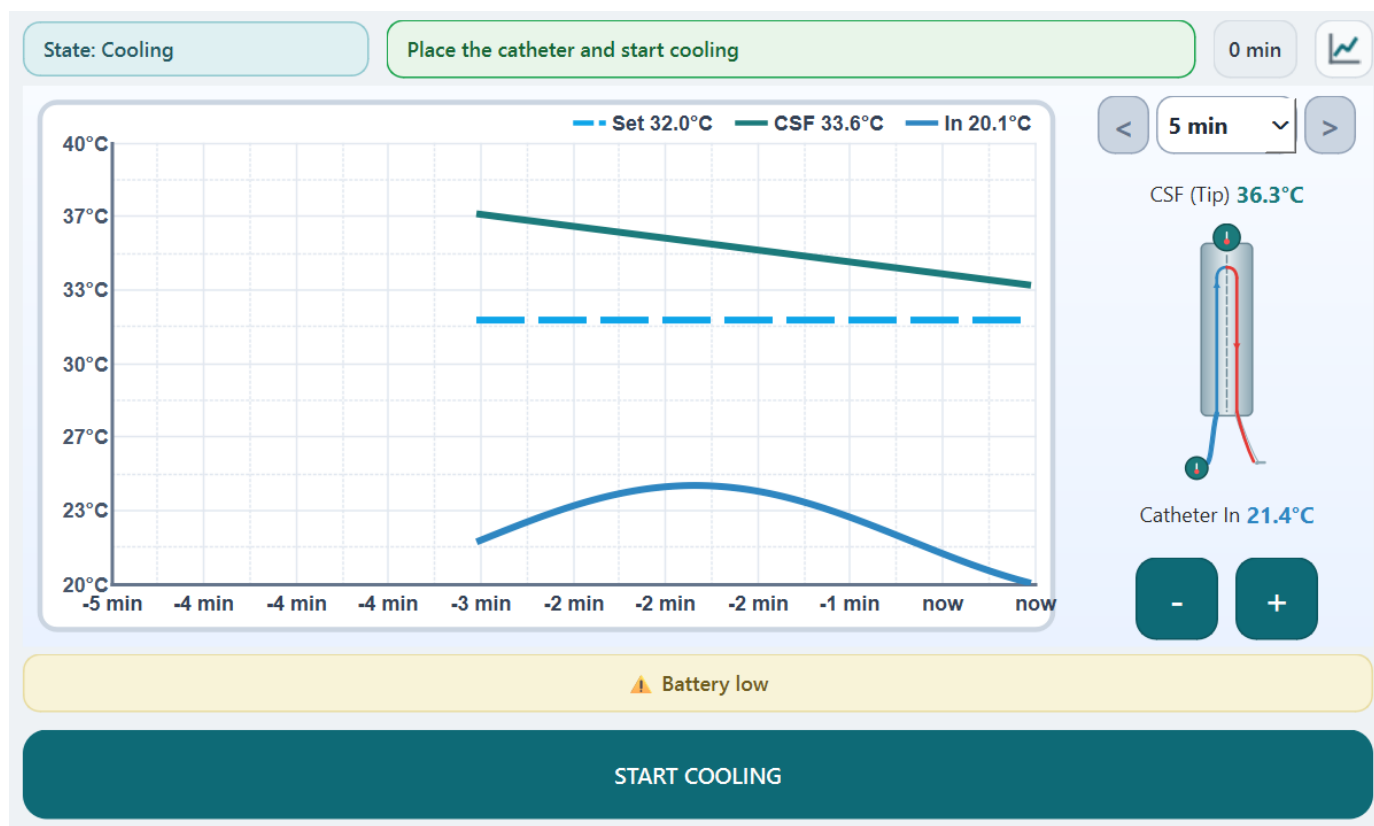
What to do

1. Inspect the cartridge fill level and top up if it is low.
2. Check for leaks around the cartridge and tubing.
3. On expert Status, confirm Level Low and Level Critical both read OK.
4. Reseat the level-sensor connectors if a switch stays off.
5. Acknowledge once both level sensors read OK.

Close

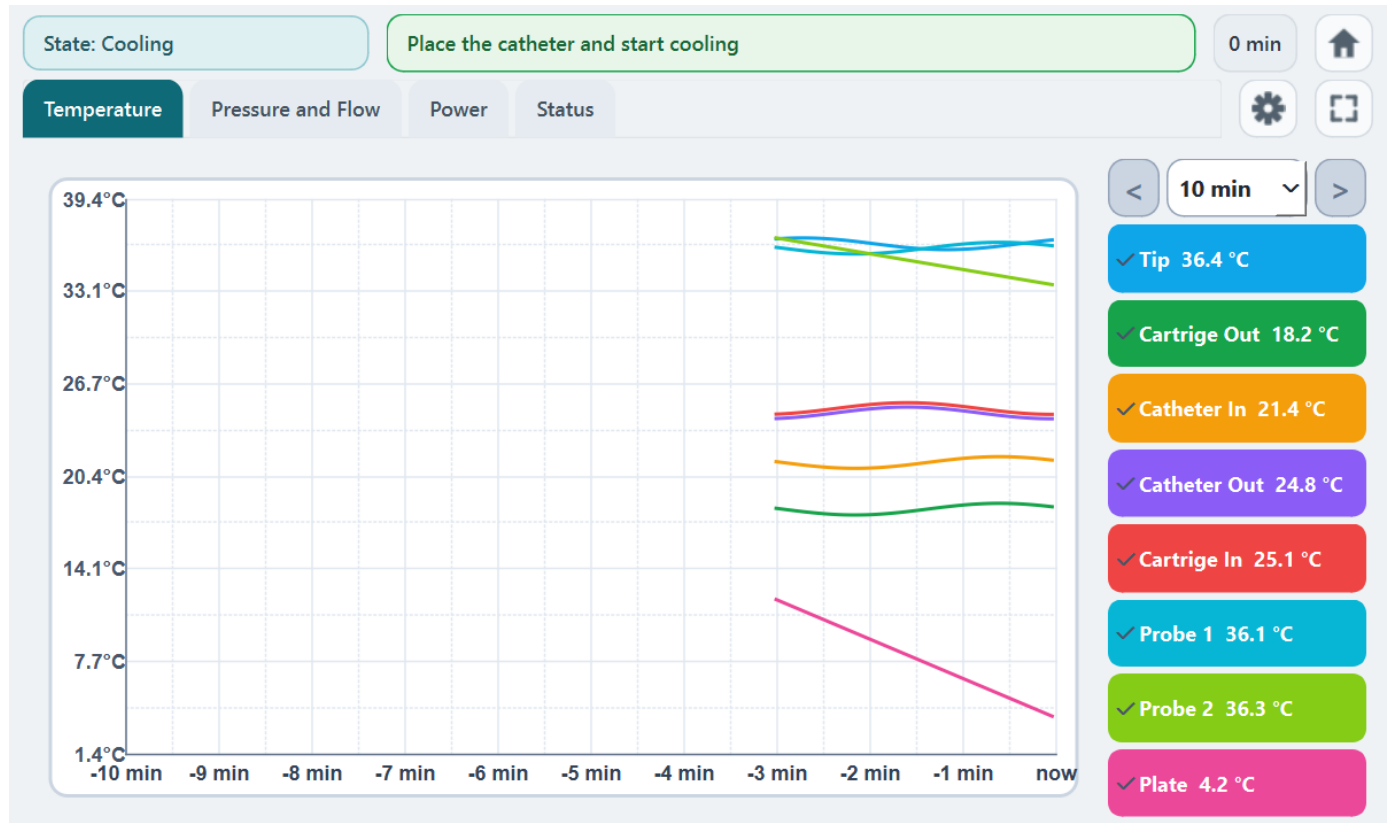
Tap the error chip to open probable causes and recovery steps. Close the sheet, fix the cause, then acknowledge.

Main screen — warning



MESSAGE-severity alerts (for example battery low) do not stop the workflow. They appear as a yellow banner until the condition clears.

Expert — Temperature



Read-only traces for all thermistors plus the setpoint. Toggle series on the right. House icon returns to the main screen.

Expert — Pressure and Flow



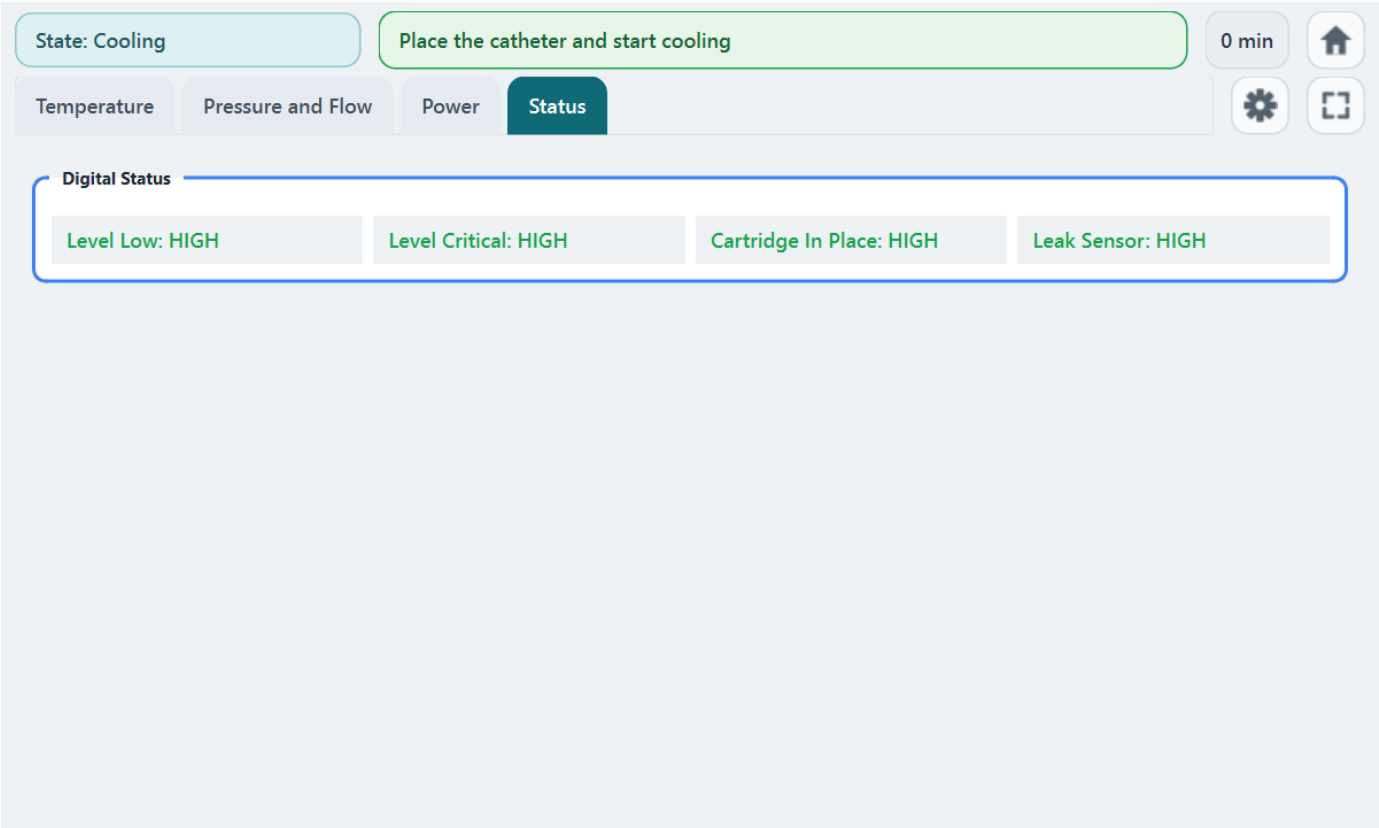
Catheter and pump pressures (psi) with pump flow (ml/min) on the right axis.

Expert — Power



Estimated catheter and cartridge cooling power from flow and ΔT . Use this to confirm heat is actually being extracted.

Expert — Status



Live digital inputs. HIGH / green means the switch is satisfied; LOW / red is a problem during cooling or pumping.

Service — Manual Operation

State: Cooling

Place the catheter and start cooling

0 min

Manual Operation

Pressure Log

Calibration

Compressor

Compressor: ON (Heat Ex 4.2°C, cooling)

Control ON

Off below (°C) - -9.0 + On above (°C) - -9.0 +

Stepper

RPM

ml/min

Calibrate 5 min

102 RPM
59.8 ml/min

JOG REV

JOG FWD

RUN

Start Flow Ramp Test

Reached from the gear icon on the expert page. Jog or run the pump, set flow, and enable compressor control. Do not use this during a patient session unless instructed.

Service — Pressure Log

State: Cooling

Place the catheter and start cooling

0 min

Manual Operation

Pressure Log

Calibration

Pressure CSV Log

Logging ON

Logging ON — writing a new pressure CSV at 100 Hz.

Toggle high-rate (100 Hz) pressure CSV capture. Each ON starts a new runNN file paired with the session temperature log.

Service — Calibration

State: Cooling

Place the catheter and start cooling

0 min

Manual Operation

Pressure Log

Calibration

2-Point Calibration

Enter measured values for each sensor (leave blank to skip a row).

Sensor	Raw (°C)	Calibrated (°C)	Measured at 0°C	Measured at 100°C
Tip	36.4	36.4	e.g. 0.2	e.g. 99.4
Cartridge Out	18.2	18.2	e.g. 0.2	e.g. 99.4
Catheter In	21.4	21.4	e.g. 0.2	e.g. 99.4
Catheter Out	24.8	24.8	e.g. 0.2	e.g. 99.4
Cartridge In	25.1	25.1	e.g. 0.2	e.g. 99.4
Probe 1	36.1	36.1	e.g. 0.2	e.g. 99.4
Probe 2	36.2	36.2	e.g. 0.2	e.g. 99.4

Apply All Calibrations

Two-point temperature calibration (measured at 0 °C and 100 °C). Apply only after a controlled ice / boiling-water check.

Error handling

STOP faults enter Error, stop the pump, and require ACKNOWLEDGE ERROR after the cause is gone. Tap the orange error chip for causes and steps. **MESSAGE** faults only show a yellow banner and do not change state. If several STOP faults are active, the highest-priority one is shown (I/O, level, cartridge, fridge, leak, CSF low, heat exchanger, cooling ineffective).

Fault	When it trips	What to do
Level sensor failure detected LEVEL_SENSOR · STOP	Cooling/Pumping and a level switch is LOW.	Inspect the cartridge fill level and top up if it is low. Check for leaks around the cartridge and tubing.
Cartridge removed during operation CARTRIDGE_REMOVED · STOP	Cooling/Pumping and cartridge switch is OFF.	Fully seat the cartridge until it is locked in place. On expert Status, confirm Cartridge In Place is ON.
CSF low temp CSF_LOW_TEMP · STOP	CSF (Probe 2) below the low-temp limit (default 28 °C).	Check the CSF temperature on the main display. Raise the setpoint if it is below the intended target.
Sensor read failure IO_READ_FAILURE · STOP	A sensor board or bus read failed.	Check sensor board power and cables. Open expert Status and confirm readings update.
Battery low BATTERY_LOW · WARN	Reported battery below 20% (warning only).	Connect mains power or replace/charge the battery. Confirm the power lead is fully seated.
Fridge defect FRIDGE_DEFECT · STOP	Compressor/fridge defect flag is set.	On Manual Operation, check whether the compressor is running. Check the plate / heat-exchanger temperature.
Leak detected LEAK_DETECTED · STOP	Leak sensor stays LOW (wet). Detection can be disabled in config.	Do not restart cooling until the area is dry. Inspect the cartridge, catheter, and tubing for leaks.
Heat exchanger too cold HEAT_EX_TOO_COLD · STOP	Plate below the heat-exchanger minimum (default -10 °C).	Check the plate / heat-exchanger temperature. Confirm the pump is circulating coolant.
Cooling ineffective COOLING_INEFFECTIVE · STOP	Pump + compressor on, but CSF does not drop enough within the timeout.	Confirm the catheter is placed and connected. Check pump speed, flow, and pressures on the expert page.